PRODUCT REFERENCE MANUAL

MEMBER OF THE UNIT DEVLAB ECOSYSTEM

UNIT PULSAR ESP32-C5

MPN: UE0134

Version 0.1.0



ESP32-C5 development board with dual-band Wi-Fi 6, Bluetooth LE, 802.15.4, environmental sensing, battery fuel gauge, microSD, QWIIC, and USB-C.

ESP32-C5HR8 | 8 MiB FLASH | 8 MB PSRAM | WI-FI 6 · BLE · 802.15.4

DOCUMENT MAP

Contents

Description

Applications
Hardware Features

1 The Board

1.1 Accessories
1.2 Board Identification
1.3 Main Assemblies
1.4 Board Views
1.5 Handling

2 Ratings

2.1 Recommended Operating Conditions
2.2 Power Domains and Power Distribution
2.3 Digital Interfaces
2.4 Confirmed Component Values
2.5 Application-Dependent and Uncharacterized Parameters
2.6 Electrical Precautions

3 Functional Overview

3.1 Block Diagram
3.2 Board Topology
3.3 Processor
3.4 Memory Architecture: Flash and PSRAM
3.5 RF Path and Antenna Connector
3.6 MicroSD Card Socket
3.7 LED Indicators
3.8 BQ24074 and SGM6029 Power Management System
3.9 Power Tree
3.10 Environmental and Magnetic Sensors
3.11 MAX17048 Fuel Gauge
3.12 I2C and QWIIC Expansion
3.13 Combined System Operation

4 Connectors & Pinouts

4.1 General Pinout
4.2 Interface Summary
4.3 Analog Inputs
4.4 Board Resources and I2C Addresses
4.5 Arduino Nano Pinout Compatibility
4.6 QWIIC Connector
4.7 J20 SPI Connector
4.8 MicroSD Connector
4.9 Battery Connections
4.10 RF Antenna Connector
4.11 USB-C, BOOT, and Reset
4.12 Debug Interface

5 Board Operation

5.1 Getting Started with Arduino IDE
5.2 First Power-up and Inspection
5.3 BOOT and Reset Workflow
5.4 Recommended Bring-up Sequence
5.5 C++ Example Collection
5.6 Memory Planning
5.7 Storage Operation
5.8 Radio, Sensor, and Storage Coexistence
5.9 Troubleshooting Guide

6 Mechanical Information

6.1 Component-side Envelope
6.2 Bottom-side Envelope
6.3 Board Dimensions
6.4 Mechanical Integration Considerations

7 Company Information

8 Reference Documentation

9 Appendix

9.1 Schematic
9.2 Technical References
9.3 Design and Application Notes
9.4 Document Control
9.6 Hardware Errata

Description

The UNIT PULSAR ESP32-C5 is a multi-interface development board in the UNIT DevLab ecosystem, built around the Espressif ESP32-C5HR8 SoC with 8 MB of in-package PSRAM. The V1.2.3 design combines dual-band 2.4/5 GHz Wi-Fi 6, Bluetooth LE, and IEEE 802.15.4 connectivity with external QSPI flash, environmental and magnetic sensing, microSD, USB-C, Li-ion charging with fuel gauging, an addressable RGB indicator, QWIIC I2C, and an RF coaxial antenna connector.

Applications

- Dual-band Wi-Fi 6, Bluetooth LE, Zigbee, and Thread prototyping
- Environmental monitoring and data logging
- Battery-powered IoT nodes with fuel gauging
- Magnetometer and proximity-based user interfaces
- microSD data logging
- Education and subsystem validation

Hardware Features

- ESP32-C5HR8 SoC: 32-bit RISC-V up to 240 MHz, 384 KB HP SRAM, 8 MB PSRAM
- BY25Q64ES 64 Mbit (8 MiB) quad SPI NOR flash
- 2.4/5 GHz RF path through a diplexer to a coaxial antenna connector
- LPS22 pressure, FHT40 humidity/temperature, VCNL4040 proximity/light, and MMC5603NJ magnetometer sensors
- BQ24074 Li-ion charger with power path and MAX17048 fuel gauge
- SGM6029 1 A buck regulator for the 3.3 V rail
- microSD socket on SPI signals
- One WS2812 RGB LED plus power, charge, and user indicators
- USB-C, QWIIC, 6-pin SPI, battery, and Nano-style edge-pad connections

1 The Board

The UNIT PULSAR ESP32-C5 is a development board for evaluating the Espressif ESP32-C5HR8 and building connected applications that combine dual-band Wi-Fi 6, Bluetooth LE, and IEEE 802.15.4 radio with environmental sensing, removable storage, battery operation, and general-purpose I/O. Schematic revision V1.2.3 places the SoC, flash, RF diplexer, power-management devices, sensors, indicators, and user controls on a Nano-style board with two parallel edge-pad rows.

1.1 Accessories

The following optional accessories are recommended for use with the UNIT PULSAR ESP32-C5. Select accessories according to the interfaces and features required by the application.

Accessory	Purpose	Selection notes
Dual-band antenna with matching coaxial plug	Wi-Fi, Bluetooth LE, and 802.15.4 radio operation	Must cover 2.4 GHz and 4.9–5.95 GHz and mate with the J2 RF1-2626 connector; no onboard antenna is fitted
QWIIC cable	External I2C expansion	4-pin, 1 mm pitch cable for connecting compatible 3.3 V QWIIC devices
USB-C data cable	Power, programming, and USB serial	Must support data; a charge-only cable cannot upload firmware
MicroSD card	Removable storage and data logging	Class 10 microSD card; FAT32 formatting is recommended
LiPo 3.7 V 650 mAh Battery	Battery-powered operation	Single-cell LiPo battery; verify polarity and connector orientation before connection

Verify connector type, orientation, polarity, voltage domain, and pinout before connecting accessories.

1.2 Board Identification

Item	Value
Product	UNIT PULSAR ESP32-C5
SKU	UE0134
Product family	UNIT DevLab ecosystem
Product type	Multi-interface ESP32-C5 development board
Main component	Espressif ESP32-C5HR8, QFN-48 (6 × 6 mm)
Hardware revision	V1.2.3 (schematic)
Product Reference	Version 0.1.0
Primary programming interface	USB-C / ESP32-C5 USB Serial/JTAG
Debug interface	USB Serial/JTAG through USB-C

Hardware revision and documentation revision are identified independently.

1.3 Main Assemblies

RefDes	Component	Function
IC1	ESP32-C5HR8	Main SoC with 8 MB in-package PSRAM and 2.4/5 GHz radio
IC3	BY25Q64ESHIG	64 Mbit (8 MiB) external quad SPI flash
IC2	BQ24074RGTR	Single-cell Li-Ion charger with power path
U2	SGM6029CYG	1 A buck regulator for the 3.3 V rail
IC4	MAX17048G+T10	Battery fuel gauge
A2	LPS22HBTR	Barometric pressure sensor
U$36	FHT40-DD-TR	Humidity and temperature sensor
U3	VCNL4040M3OE	Proximity and ambient-light sensor
U4	MMC5603NJ	Three-axis magnetometer
U1	LFD182G45DCHD277	2.4 GHz / 5 GHz RF diplexer
J2	RF1-2626-125CE-CT	RF coaxial antenna connector
XTAL1	SX0B48.000F0810F30	48 MHz main crystal
MICRO_SD-HOLDER	47309-2651	Removable microSD storage
LED1	WS2812 1010	Addressable RGB indicator
J1	HCZZ0032-4	Four-position QWIIC-style I2C connector
J20	SM06B-SRSS-TB	Six-position, 1 mm SPI expansion connector
JP1, J3	PH 2.0 mm 2P, 1.25 mm 2P	Parallel battery connections
S1, S2	1TS026A-1600-0553-CT	BOOT and RESET push buttons

1.4 Board Views

Top-side and bottom-side board photographs for revision V1.2.3 are not yet available. Refer to the schematic in Section 9.1 for component designators.

1.5 Handling

Handle the board using normal ESD precautions. Avoid touching the RF connector, sensor apertures, connector contacts, or exposed pads. Remove power before connecting or removing the battery. Connect an antenna to J2 before enabling radio transmission. Keep conductive objects away from the battery and microSD areas when the board is energized.

2 Ratings

This chapter defines the recommended operating conditions and electrical characteristics of the UNIT PULSAR ESP32-C5 hardware V1.2.3. Ratings apply to the complete board unless otherwise stated.

2.1 Recommended Operating Conditions

The UNIT PULSAR ESP32-C5 can be powered from USB-C, VIN, or a single-cell LiPo battery. The BQ24074 power-path charger feeds the onboard SGM6029 buck regulator, which supplies the 3.3 V logic domain used by the ESP32-C5HR8 and onboard peripherals.

Parameter	Recommended operating condition	Notes
USB-C input	5 V nominal	Primary power and programming interface
VIN input	5 V nominal	Joins USB VBUS through Schottky diode D4; board-level range not characterized
Battery input	3.7 V nominal, 4.2 V maximum	Single-cell LiPo battery
Regulated logic rail	3.3 V nominal	Generated by the onboard SGM6029 buck regulator
3.3 V regulator output capability	Up to 1 A	SGM6029 component rating; shared by the SoC, peripherals, LEDs, and external 3.3 V loads
Battery charge current	250 mA nominal	Set by R13 (3.57 kΩ) on the BQ24074 `ISET` pin
Input current limit	≈1.4 A	Set by R12 (1.1 kΩ) on the BQ24074 `ILIM` pin
ESP32-C5 CPU clock	Up to 240 MHz	Firmware configurable
Main crystal	48 MHz	XTAL1
Internal SRAM	384 KB HP SRAM, 16 KB LP SRAM	ESP32-C5 internal memory
External flash	64 Mbit / 8 MiB	BY25Q64ESHIG
PSRAM	8 MB	In-package, ESP32-C5HR8
Logic level	3.3 V	Applies to onboard digital interfaces and GPIO

2.2 Power Domains and Power Distribution

The UNIT PULSAR ESP32-C5 uses separate power domains for external supply, battery operation, system power, and regulated 3.3 V logic.

The main power domains are:

- **USB-C / VBUS:** provides the nominal 5 V USB supply used for board power, programming, and battery charging. VBUS reaches the +5V node through Schottky diode D1 (SDM2U30CSP-7B, 2 A).
- **VIN:** external supply input that reaches the +5V node through Schottky diode D4 (SDM2U30CSP-7B, 2 A).

- **+5V:** input of the BQ24074 charger and power-path controller.
- **VBAT:** battery domain intended for a single-cell LiPo battery with a nominal voltage of 3.7 V and a maximum fully charged voltage of 4.2 V.
- **BMS_OUT:** BQ24074 power-path output used as the system supply and as the input of the 3.3 V buck regulator.
- **3.3 V:** regulated logic supply generated by the SGM6029. This rail powers the ESP32-C5HR8, external flash, sensors, fuel gauge, microSD interface, QWIIC and J20 connectors, RGB LED, and indicators.

The SGM6029 is rated for up to 1 A output current. This current is shared between the ESP32-C5HR8, onboard peripherals, LEDs, and any external device powered from the 3.3 V rail. Radio transmission produces short current peaks, so the current available to external loads depends on the active board configuration and application workload.

The BQ24074RGTR manages single-cell LiPo battery charging. The schematic annotates a nominal charge current of 250 mA (`ICHRG=250mA`), set by R13. The input current limit is set by R12. Both values may be changed by modifying the corresponding resistor according to the BQ24074 configuration requirements.

All onboard digital interfaces operate in the 3.3 V logic domain. QWIIC, J20, microSD, and exposed GPIO signals must therefore be used with 3.3 V-compatible devices unless external level translation is provided.

Connector mechanical current ratings must not be interpreted as board power ratings. Maximum usable current is also constrained by the associated PCB traces, power path, regulator, protection components, and operating conditions.

2.3 Digital Interfaces

All onboard digital peripherals are connected to the board's 3.3 V logic domain. External logic compatibility must nevertheless be checked against the ESP32-C5 input/output specifications and the power state of both devices.

Interface	Board connection	Electrical note
Sensor I2C	GPIO2 / GPIO3	Onboard sensors and fuel gauge; 4.7 kΩ pull-ups; strapping pins
QWIIC I2C	GPIO1 / GPIO6	Connector also supplies 3.3 V and GND
SPI / microSD	GPIO27, GPIO7, GPIO26, GPIO23	SCK, MOSI, MISO, microSD CS; shared with J20
J20 chip select	GPIO10	SS for an external SPI device
WS2812 data	GPIO0	Drives the onboard RGB LED
UART0	GPIO11 / GPIO12	TX through 499 Ω series resistor
USB	GPIO13 / GPIO14	Native USB Serial/JTAG

2.4 Confirmed Component Values

Symbol / parameter	Value	Scope
ESP32-C5 HP SRAM	384 KB	ESP32-C5HR8 component
ESP32-C5 CPU clock	240 MHz maximum	ESP32-C5HR8 component

Symbol / parameter	Value	Scope
In-package PSRAM	8 MB	ESP32-C5HR8 component
QSPI flash capacity	64 Mbit	IC3
Buck output current	1 A	U2 component rating
Charge current	250 mA	R13 = 3.57 kΩ
Crystal frequency	48 MHz	XTAL1
Sensor I2C pull-ups	4.7 kΩ	R15, R16

2.5 Application-Dependent and Uncharacterized Parameters

Some electrical and performance characteristics depend on firmware configuration, operating mode, connected peripherals, and application load. Unless otherwise specified, these parameters should not be interpreted as guaranteed board-level limits.

- **Available 3.3 V current for external loads:** the SGM6029 provides up to 1 A total output current. This current is shared by the ESP32-C5HR8, radio transmission peaks, flash, sensors, LEDs, and external 3.3 V loads.
- **Board current consumption:** total current consumption varies with CPU activity, radio mode and transmit power, enabled peripherals, RGB LED brightness, microSD operation, and connected external devices.
- **RF performance:** radiated performance depends on the external antenna, cable, enclosure, and the populated matching network. It is not characterized at board level.
- **Low-battery operation:** the SGM6029 input is the BQ24074 system output. 3.3 V regulation at low battery voltage is not characterized at board level.
- **GPIO electrical characteristics:** GPIO drive strength, input thresholds, and other pin-level electrical characteristics follow the ESP32-C5 specifications unless a board-level circuit modifies the corresponding signal.
- **Environmental characteristics:** board-level operating temperature, humidity, ESD immunity, and thermal performance are not specified unless explicitly stated in a released qualification or validation specification.

2.6 Electrical Precautions

1. Use only the specified power inputs: 5 V nominal through USB-C or VIN, or a 3.7 V nominal / 4.2 V maximum single-cell LiPo battery.
2. All GPIO and onboard digital interfaces operate in the 3.3 V logic domain. **Do not apply 5 V directly to GPIO, QWIIC, J20, microSD, or other 3.3 V signal lines.**
3. Observe the battery connector polarity before connecting a single-cell LiPo battery. Do not connect multi-cell battery packs or batteries exceeding 4.2 V. Do not connect batteries to JP1 and J3 at the same time; both connectors are wired in parallel.
4. Do not externally drive the 3.3 V rail while the onboard regulator is active unless the external power source and operating configuration have been specifically designed for this purpose.
5. Connect external circuits to a common ground before applying external clock, data, or control signals.
6. Do not hold GPIO2, GPIO3, GPIO7, GPIO25, GPIO26, GPIO27, or GPIO28 at a level that changes the boot configuration during reset; these are ESP32-C5 strapping pins.

7. Connect an antenna to J2 before enabling radio transmission.
8. Ensure that the total load on the 3.3 V rail, including onboard circuitry and externally connected devices, remains within the 1 A capability of the SGM6029 regulator.

3 Functional Overview

The UNIT PULSAR ESP32-C5 is organized around the ESP32-C5HR8 SoC, with an external flash, a dual-band RF path, fixed onboard sensors, and a battery power path. Firmware can use each block independently or combine wireless connectivity, sensing, storage, and user feedback in one application.

3.1 Block Diagram

A block-diagram drawing for revision V1.2.3 is not yet available. The functional relationships are:

```
USB-C / VIN / battery
          |
   BQ24074 power path ---- MAX17048 fuel gauge
          |
   SGM6029 buck -> 3.3 V
          |
     ESP32-C5HR8 (8 MB PSRAM in package)
   +------+--------+--------+-------+---------+
   |      |        |        |       |         |
 QSPI    RF      I2C       SPI    WS2812    GPIO
 flash  diplexer  sensors  microSD  RGB     + QWIIC
        + J2               + J20
```

The ESP32-C5HR8 controls every data path. The BY25Q64 flash stores executable code and nonvolatile application data. The in-package PSRAM extends working memory. The microSD socket provides removable storage. The pressure, humidity/temperature, proximity/light, and magnetic sensors and the battery fuel gauge share an internal I2C bus. QWIIC, J20, and the edge pads expose expansion signals, while the RF connector and RGB LED provide wireless and visual outputs.

3.2 Board Topology

RefDes	Component	Confirmed role
IC1	ESP32-C5HR8	Main SoC with in-package 8 MB PSRAM
IC3	BY25Q64ESHIG	64 Mbit (8 MiB) QSPI flash memory
IC2	BQ24074RGTR	Single-cell Li-Ion/LiPo charger with power path
U2	SGM6029CYG	4 MHz, 1 A buck regulator for the 3.3 V rail
L2	470 nH inductor	SGM6029 output inductor
IC4	MAX17048G+T10	Battery fuel gauge
A2	LPS22HBTR	Barometric pressure sensor
U$36	FHT40-DD-TR	Humidity and temperature sensor
U3	VCNL4040M3OE	Proximity and ambient-light sensor
U4	MMC5603NJ	Three-axis magnetometer
U1	LFD182G45DCHD277	2.4 GHz / 5 GHz RF diplexer
J2	RF1-2626-125CE-CT	RF coaxial antenna connector

RefDes	Component	Confirmed role
XTAL1	48 MHz crystal	ESP32-C5 main clock source
MICRO_SD-HOLDER	47309-2651	microSD card socket on SPI signals
LED1	WS2812 1010	Addressable RGB LED
J1	HCZZ0032-4	Four-position, 1 mm pitch QWIIC connector
J20	SM06B-SRSS-TB	Six-position, 1 mm pitch SPI connector
NANO1	USB-C connector	USB power, programming, and USB data interface
JP1, J3	Battery connectors	PH 2.0 mm and 1.25 mm, wired in parallel
S1	BOOT switch	Pulls GPIO28 low to enter download boot mode
S2	RESET switch	Pulls CHIP_PU low
D1, D4	SDM2U30CSP-7B	Schottky diodes from VBUS and VIN to +5V

The board also provides solder jumpers that connect optional signals:

Solder jumper function	Connection
3.3 V regulator enable	SGM6029 EN to header position 3 (EN)
Battery access	VBAT to header position 4
RGB chain extension	LED1 DO to header position 11
Ground	GND to header position 13 (Nano RESET position)
Sensor bus	SDA_SENSOR / SCL_SENSOR to GPIO2 / GPIO3
Fuel-gauge alert	MAX17048 ALERT to D5 / GPIO23
Fuel-gauge quick start	MAX17048 QSTRT to D6 / GPIO24
Charger power good	BQ24074 PGOOD to D7 / GPIO25
Proximity interrupt	VCNL4040 INT to A6 / GPIO6

The default state of each solder jumper is not stated in the schematic; verify it on the assembled board.

The two edge-pad rows expose power, analog, serial, SPI, and general-purpose signals. Several of these signals are shared with onboard devices, so firmware ownership must be decided before reusing them.

3.3 Processor

The ESP32-C5HR8 is the main controller. According to the Espressif component datasheet, the device provides a high-performance 32-bit RISC-V core up to 240 MHz, a low-power RISC-V core up to 48 MHz, 384 KB of HP SRAM, 16 KB of LP SRAM, and 320 KB of ROM.

The radio supports dual-band 2.4 GHz and 5 GHz Wi-Fi 6 (802.11ax) together with 802.11a/b/g/n, Bluetooth LE (Bluetooth Core 6.0 certified), and IEEE 802.15.4 for Thread 1.4 and Zigbee 3.0.

The peripheral set includes a 12-bit SAR ADC with up to six channels, UART, I2C, I2S, a general-purpose SPI port, USB Serial/JTAG, two CAN FD controllers, LED PWM, MCPWM, RMT, PARLIO, and GDMA. The PULSAR design routes a selected subset of these functions to onboard devices and external connectors.

The ESP32-C5 does not include an SD host controller; the microSD socket is therefore wired for SPI mode.

3.4 Memory Architecture: Flash and PSRAM

The BY25Q64ESHIG provides 64 Mbit (8 MiB) of external quad SPI flash on the dedicated flash pins (GPIO15–GPIO18 and GPIO20–GPIO22). It stores the bootloader, partition table, application firmware, and any filesystem or data region defined by the selected partition scheme. These pins are not available for other uses.

The ESP32-C5HR8 includes 8 MB of quad SPI PSRAM inside the package. In the Arduino workflow, PSRAM must be enabled in the board options; ESP-IDF enables it through `menuconfig`. PSRAM is suited to network buffers, file buffers, sensor history, and large application structures. Interrupt state and timing-critical data should remain in internal SRAM. The application must check allocation results.

3.5 RF Path and Antenna Connector

The ESP32-C5HR8 provides separate RF ports for the 2.4 GHz band (`ANT_2G`, pin 42) and the 5 GHz band (`ANT_5G`, pin 48). The U1 LFD182G45DCHD277 diplexer combines both ports into a single path covering 2.4–2.5 GHz and 4.9–5.95 GHz, which is routed to the J2 RF1-2626-125CE-CT coaxial connector.

Path	Series element	Shunt positions
J2 to diplexer common	L5 (0 Ω)	C20, C21 (not populated)
Diplexer LF to `ANT_2G`	L1 (0 Ω)	C7, C9 (not populated)
Diplexer HF to `ANT_5G`	L3 (0 Ω)	C18, C4 (not populated)

The 0 Ω links and unpopulated capacitor positions form tuning networks that can be adjusted for a specific antenna. No onboard antenna is fitted. Connect a dual-band antenna to J2 before enabling radio transmission.

3.6 MicroSD Card Socket

The onboard 47309-2651 microSD socket is connected in SPI mode and operates in the 3.3 V logic domain.

Socket signal	GPIO	SPI role
`CLK`	27	SCK
`CMD`	7	MOSI
`DAT0`	26	MISO
`DAT3`	23	Chip select
`DAT1`, `DAT2`	—	Not connected

SCK, MOSI, and MISO are shared with the J20 connector and the D11–D13 header positions. External SPI devices must use their own chip select, such as GPIO10 on J20. GPIO23 is also the optional MAX17048 ALERT input.

File writes should be flushed and closed before card extraction or power removal to reduce the risk of filesystem corruption. FAT32 is recommended.

3.7 LED Indicators

LED1 is a WS2812 1010 addressable RGB LED driven by GPIO0 (NEOP_DIN), which is also available at header position 20. A solder jumper can connect LED1 D0 to header position 11, allowing the chain to be extended with compatible external addressable LEDs.

The board also includes a user indicator (BUILTIN2, pink) on GPIO27 through a 10 kΩ resistor, a red power indicator on the 3.3 V rail, and an orange charge-status indicator driven by the BQ24074 CHG output.

Applications should limit RGB brightness when power consumption or thermal rise is important.

3.8 BQ24074 and SGM6029 Power Management System

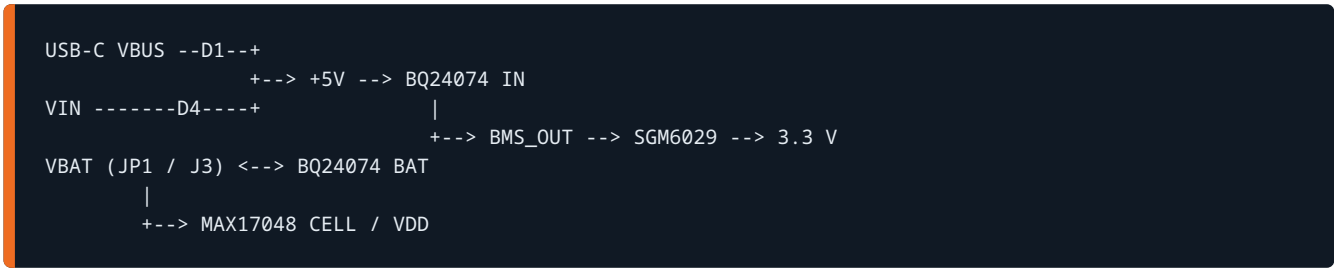
IC2 is a BQ24074RGTR single-cell Li-Ion/LiPo charger with dynamic power-path management. Its IN pin is fed from the +5V node, which combines USB VBUS (through D1) and VIN (through D4). Its OUT pin (BMS_OUT) supplies the system while the battery charges.

BQ24074 pin	Board configuration
ISET	R13 = 3.57 kΩ; 250 mA nominal charge current
ILIM	R12 = 1.1 kΩ; ≈1.4 A input current limit
EN1, EN2	GND and BMS_OUT; input limit set by ILIM
CE	GND; charging enabled
TS	10 kΩ to GND (R11); no battery thermistor
CHG	Orange charge LED
PGOOD	Optional solder jumper to D7 / GPIO25

U2 is an SGM6029CYG 4 MHz synchronous buck regulator. It converts BMS_OUT into the regulated 3.3 V rail, with the output voltage selected by R14 on the VSEL/MODE pin. The regulator is rated for up to 1 A. Its EN pin is pulled up to BMS_OUT by R8 (10 kΩ) and can be connected to header position 3 through a solder jumper for external control of the 3.3 V rail.

The battery input is intended for a single-cell LiPo battery with a nominal voltage of 3.7 V and a maximum fully charged voltage of 4.2 V. JP1 (PH 2.0 mm) and J3 (1.25 mm) are wired in parallel to VBAT.

3.9 Power Tree



The power tree summarizes the main supply domains and their relationship. It is intended as a functional reference and does not replace the complete schematic.

The 3.3 V rail is available through the edge pads, the QWIIC connector, and J20. The total 3.3 V load, including radio transmission peaks, must remain within the 1 A regulator output capability.

3.10 Environmental and Magnetic Sensors

Four sensors share the internal SDA_SENSOR / SCL_SENSOR I2C bus with the fuel gauge. R15 and R16 (4.7 kΩ) pull the bus up to 3.3 V, and solder jumpers connect it to GPIO2 (SDA) and GPIO3 (SCL).

Device	Measurement	I2C address	Notes
LPS22HBTR (A2)	Barometric pressure, 260–1260 hPa	0x5C	SA0 tied to GND; INT_DRDY not connected
FHT40-DD-TR (U$36)	Relative humidity and temperature	0x44	SHT40-compatible default; verify for the assembled part
VCNL4040M3OE (U3)	Proximity and ambient light	0x60	INT to GPIO6 through optional solder jumper
MMC5603NJ (U4)	Three-axis magnetic field	0x30	Compass and orientation applications

Typical applications include weather stations, indoor air monitoring, altitude estimation, gesture or presence detection, automatic display brightness, and compass headings. Measurement range, filtering, output data rate, and interrupt behavior are configured through the sensor registers or software driver and are not fixed by the PCB.

3.11 MAX17048 Fuel Gauge

IC4 is a MAX17048G+T10 single-cell fuel gauge with ModelGauge. Its CELL and VDD pins connect to VBAT, and it communicates on the sensor I2C bus at address 0x36. Firmware can read battery voltage and state of charge without an external sense resistor.

The ALERT output can be connected to D5 / GPIO23 and the QSTRT input to D6 / GPIO24 through optional solder jumpers. Because GPIO23 is also the microSD chip select, applications using both functions must account for this shared connection before closing the ALERT jumper.

3.12 I2C and QWIIC Expansion

The UNIT PULSAR ESP32-C5 provides two independent I2C routes, allowing the onboard sensors and external QWIIC devices to operate on separate buses.

Interface	SDA	SCL	Primary use
Sensor I2C	GPIO2	GPIO3	Onboard sensors, fuel gauge, and edge-pad access
QWIIC I2C	GPIO1	GPIO6	External QWIIC devices

GPIO2 and GPIO3 are also exposed at the SDA / SCL edge pads and are ESP32-C5 strapping pins (MTMS, MTDI). GPIO1 and GPIO6 are also exposed at the A1 and A6 edge pads.

Both interfaces operate in the 3.3 V logic domain. Device addresses, pull-up configuration, bus capacitance, and cable length must be considered when multiple devices are connected to the same bus.

3.13 Combined System Operation

The UNIT PULSAR ESP32-C5 is designed to support concurrent operation of its onboard subsystems. The ESP32-C5HR8 can combine wireless connectivity, environmental sensing, battery monitoring, removable storage, and RGB indication within the same application.

For example, sensor data can be buffered in PSRAM, logged to the microSD card, and published over Wi-Fi or Thread, while the fuel gauge reports battery state and the RGB LED indicates connection status.

Applications using the radio together with storage and sensors should consider current peaks during transmission, memory allocation, task priorities, and storage write latency. Shared GPIO functions must also be considered when enabling optional solder-jumper connections.

4 Connectors & Pinouts

Pin labels describe hardware revision V1.2.3. GPIO numbers identify ESP32-C5HR8 signals according to the Espressif datasheet pin table; framework aliases depend on the selected board definition.

4.1 General Pinout

Pos.	Board label	ESP32-C5HR8	Primary board role	Shared resource
1	D13 / SCK / LED	GPIO27	SPI clock	microSD CLK, J20, BUILTIN2 LED; strapping pin
2	3V3	—	3.3 V rail	SGM6029 output
3	EN	—	3.3 V regulator enable	Solder jumper to SGM6029 EN
4	VBAT	—	Battery rail	Solder jumper to VBAT
5	A1 / D15	GPIO1 / ADC1_CH0	Analog-capable I/O	QWIIC SDA
6	A2 / D16	GPIO4 / ADC1_CH3	Analog-capable I/O	MTCK
7	A3 / D17	GPIO5 / ADC1_CH4	Analog-capable I/O	MTDO
8	SDA / D18	GPIO2 / ADC1_CH1	I2C data	Sensor bus; MTMS strapping pin
9	SCL / D19	GPIO3 / ADC1_CH2	I2C clock	Sensor bus; MTDI strapping pin
10	A6 / D21	GPIO6 / ADC1_CH5	Analog-capable I/O	QWIIC SCL; optional VCNL4040 INT
11	NEOPIXEL DOUT	—	RGB-chain extension output	Solder jumper to LED1 DO
12	VBUS	—	USB VBUS	USB-C supply
13	GND	—	Ground	Solder jumper (Nano RESET position)
14	GND	—	Ground	Common return
15	VIN	—	Supply input	To +5V through D4
16	TX0 / D1	GPIO11	UART0 TX	499 Ω series resistor R3
17	RX0 / D0	GPIO12	UART0 RX	General-purpose
18	RST	CHIP_PU	Reset	RESET button, 10 kΩ pull-up
19	GND	—	Ground	Common return
20	D2 / RGB	GPIO0	NeoPixel data	LED1 DI
21	D3 / BOOT	GPIO28	BOOT input	BOOT button; strapping pin
22	D4	GPIO9	Digital I/O	General-purpose

Pos.	Board label	ESP32-C5HR8	Primary board role	Shared resource
23	D5	GPIO23	Digital I/O	microSD CS; optional MAX17048 ALERT
24	D6	GPIO24	Digital I/O	Optional MAX17048 QSTRT
25	D7	GPIO25	Digital I/O	Optional BQ24074 PG00D; strapping pin
26	D8	USB D-	USB D-	Shared with USB-C; GPIO13 through 22 Ω R9
27	D9	USB D+	USB D+	Shared with USB-C; GPIO14 through 22 Ω R10
28	D10 / SS	GPIO10	SPI chip select	J20 SS
29	D11 / MOSI	GPIO7	SPI MOSI	microSD CMD, J20; strapping pin
30	D12 / MISO	GPIO26	SPI MISO	microSD DAT0, J20; strapping pin

EN, VBAT, NEOPIXEL DOUT, and the ground at position 13 are connected to their edge-pad positions through solder jumpers.

4.2 Interface Summary

Interface	Signal	Board label	GPIO	Notes
UART0	RX	D0	GPIO12	
UART0	TX	D1	GPIO11	499 Ω series resistor R3
SPI	SS / CS	D10	GPIO10	Also J20 pin 4
SPI	MOSI	D11	GPIO7	Shared with microSD and J20
SPI	MISO	D12	GPIO26	Shared with microSD and J20
SPI	SCK	D13	GPIO27	Shared with microSD, J20, and BUILTIN2 LED
microSD	CS	D5	GPIO23	Separate chip select from D10
Sensor I2C (SDA_SENSOR)	SDA	SDA / D18	GPIO2	Onboard sensors through solder jumper; 4.7 kΩ pull-up
Sensor I2C (SCL_SENSOR)	SCL	SCL / D19	GPIO3	Onboard sensors through solder jumper; 4.7 kΩ pull-up
QWIIC I2C	SDA	A1 / D15	GPIO1	J1 pin 3; no onboard pull-up

Interface	Signal	Board label	GPIO	Notes
QWIIC I2C	SCL	A6 / D21	GPIO6	J1 pin 4; no onboard pull-up
USB	D-	D8	GPIO13	Native USB; 22 Ω R9
USB	D+	D9	GPIO14	Native USB; 22 Ω R10

The SDA / SCL edge pads are the sensor bus, not a separate external bus: any device connected there shares the bus with the five onboard I2C devices. QWIIC is an independent bus on GPIO1 and GPIO6.

Two schematic net labels differ from the connections drawn at the header:

- **D2:** IC1 GPIO8 carries the net label D2, but header position 20 (D2) is wired to NEOP_DIN / GPIO0. No other connection is shown for the GPIO8 net; treat GPIO8 as unassigned until the PCB is verified.
- **A6:** the net is named A6/GPIO4/TOUCH4, but it connects to IC1 pin 15, which is GPIO6. GPIO4 is A2 / D16 (pin 13). The ESP32-C5 has no touch sensor, so TOUCH4 does not apply.

4.3 Analog Inputs

Analog label	Digital label	GPIO	ADC channel	Shared use
A1	D15	GPIO1	ADC1_CH0	QWIIC SDA
A2	D16	GPIO4	ADC1_CH3	—
A3	D17	GPIO5	ADC1_CH4	—
A4 / SDA	D18	GPIO2	ADC1_CH1	Sensor I2C SDA
A5 / SCL	D19	GPIO3	ADC1_CH2	Sensor I2C SCL
A6	D21	GPIO6	ADC1_CH5	QWIIC SCL; optional VCNL4040 INT

The Nano A0 and A7 positions carry VBAT and NEOPIXEL DOUT and are not analog inputs. A4/A5 and A1/A6 are usable as analog inputs only while the corresponding I2C bus is not in use.

4.4 Board Resources and I2C Addresses

Resource	Signal / GPIO	Bus	I2C address	Notes
Native USB	GPIO13 / GPIO14	USB Serial/JTAG	—	D8 / D9
BOOT button	GPIO28	—	—	D3; strapping pin
UART0	GPIO11 / GPIO12	UART	—	D1 / D0
NeoPixel in	GPIO0 (NEOP_DIN)	—	—	LED1 WS2812 1010; D2
NeoPixel out	LED1 DO (NEOP_DO)	—	—	Header position 11 through solder jumper
Flash	GPIO15–GPIO18, GPIO20–GPIO22	SPI0/1	—	BY25Q64; not available as GPIO
PSRAM	In package	SPI0/1	—	8 MB, ESP32-C5HR8

Resource	Signal / GPIO	Bus	I2C address	Notes
microSD	GPIO27, GPIO7, GPIO26, GPIO23	SPI	—	CS on D5
QWIIC	GPIO1 / GPIO6	QWIIC I2C	—	J1
Battery monitor	GPIO2 / GPIO3	Sensor I2C	`0x36`	MAX17048G+T10
Light / proximity	GPIO2 / GPIO3	Sensor I2C	`0x60`	VCNL4040M3OE
Temperature / RH	GPIO2 / GPIO3	Sensor I2C	`0x44`	FHT40-DD-TR (SHT40 in schematic); verify address for FHT40
Pressure	GPIO2 / GPIO3	Sensor I2C	`0x5C`	LPS22HBTR (LPS22DFTR in schematic); `SA0` to GND
Magnetometer	GPIO2 / GPIO3	Sensor I2C	`0x30`	MMC5603NJ
RF	`ANT_2G` (pin 42), `ANT_5G` (pin 48)	—	—	U1 diplexer to J2

4.5 Arduino Nano Pinout Compatibility

The board uses two parallel 15-position edge rows based on the Arduino Nano layout. This provides a familiar mechanical footprint and labeling pattern, but does not imply full electrical or shield compatibility.

Important differences include:

- The ESP32-C5HR8 and board GPIO operate in the 3.3 V logic domain.
- Several conventional positions have PULSAR-specific functions: `EN` at `AREF`, `VBAT` at `A0`, `NEOPIXEL DOUT` at `A7`, VBUS at 5V, and a jumper-selected ground at the second RESET position.
- D2 carries the RGB LED data and D3 carries the BOOT signal.
- D8 and D9 carry the ESP32-C5 native USB D-/D+ signals and should not be treated as conventional digital I/O.
- GPIO2, GPIO3, GPIO7, GPIO25, GPIO26, GPIO27, and GPIO28 are strapping pins. External circuits must not force a level that changes the boot mode during reset.
- Analog channel labels do not follow a sequential GPIO order.

Before installing the board into a Nano-style carrier, verify power, reset, analog, digital, and mechanically shared positions.

4.6 QWIIC Connector

J1 is a four-position, 1 mm-pitch QWIIC connector carrying:

Pin	Signal	ESP32-C5HR8 connection	Function
1	GND	Ground	Common return
2	3.3 V	Regulated rail	Peripheral supply
3	SDA	GPIO1	External I2C data
4	SCL	GPIO6	External I2C clock

The QWIIC interface operates in the 3.3 V logic domain and is separate from the GPIO2/GPIO3 sensor bus. The board does not fit pull-up resistors on GPIO1/GPIO6; QWIIC devices normally provide them.

The QWIIC 3.3 V supply is provided by the SGM6029 regulator, which supports up to 1 A total output current shared between the onboard circuitry and all external 3.3 V loads.

4.7 J20 SPI Connector

J20 is a six-position, 1 mm-pitch SH connector for external SPI devices.

Pin	Signal	ESP32-C5HR8 connection
1	SCK	GPIO27
2	MOSI	GPIO7
3	MISO	GPIO26
4	SS	GPIO10
5	3.3 V	Regulated rail
6	GND	Ground

SCK, MOSI, and MISO are shared with the microSD socket. Keep the microSD chip select (GPIO23) high while communicating with a J20 device.

4.8 MicroSD Connector

Socket signal	GPIO	Description
CLK	GPIO27	SPI SCK
CMD	GPIO7	SPI MOSI
DAT0	GPIO26	SPI MISO
DAT3	GPIO23	SPI chip select
DAT1, DAT2	—	Not connected
VDD	3.3 V	Card supply
GND / shield	Ground	Electrical return and connector shield

Insert or remove the microSD card only when filesystem activity has stopped. Software should flush and close open files before card extraction or power removal.

4.9 Battery Connections

JP1 (PH 2.0 mm) and J3 (1.25 mm) provide two battery connector options wired in parallel to VBAT. Connect only one battery at a time.

The battery input is designed for a single-cell LiPo battery with a nominal voltage of 3.7 V and a maximum fully charged voltage of 4.2 V. Charging is managed by the onboard BQ24074RGTR with a nominal charge current of 250 mA, and battery state is reported by the MAX17048 fuel gauge.

Always verify connector polarity before connecting a battery, as connector compatibility alone does not guarantee correct polarity.

A solder jumper can connect the battery rail to edge-pad position 4 (VBAT).

Do not connect multi-cell battery packs or batteries exceeding 4.2 V.

4.10 RF Antenna Connector

J2 is an RF1-2626-125CE-CT coaxial connector rated to 6 GHz. It carries the combined 2.4 GHz and 5 GHz signal from the U1 diplexer. Use a dual-band antenna with a mating plug and connect it before enabling radio transmission.

4.11 USB-C, BOOT, and Reset

The USB-C connector provides power, programming, and USB data connectivity. The USB D- and D+ signals connect to the ESP32-C5 USB Serial/JTAG controller (GPIO13 and GPIO14) through 22 Ω series resistors and are also exposed at the D8 and D9 positions. CC1 and CC2 have 5.1 kΩ pull-down resistors.

The BOOT button pulls GPIO28 low. Holding BOOT during reset selects the ESP32-C5 download boot mode used for firmware programming and recovery. The RESET button pulls CHIP_PU low and restarts the ESP32-C5 while board power remains applied.

A typical firmware-recovery sequence is to hold BOOT, press and release RESET, then release BOOT after the device enumerates in download mode.

4.12 Debug Interface

The ESP32-C5 USB Serial/JTAG controller provides serial console, firmware download, and JTAG debugging through the USB-C connector without an external probe. The JTAG pins (MTMS, MTDI, MTCK, MTDO) are used as GPIO2–GPIO5 on this board and are not routed to dedicated debug pads.

5 Board Operation

The procedures below define the board workflow for the Arduino IDE and the Espressif Arduino core.

5.1 Getting Started with Arduino IDE

1. Install Arduino IDE 2.x.
2. Add the Espressif package index to **Additional Boards Manager URLs**: `https://espressif.github.io/arduino-esp32/package_esp32_index.json`
3. Install the **esp32 by Espressif Systems** package, version 3.3 or later (ESP32-C5 support).
4. Select **ESP32C5 Dev Module** or an installed PULSAR-specific board definition.
5. Set **Flash Size** to 8 MB, enable **PSRAM**, and enable **USB CDC On Boot** to use the USB-C serial port.
6. Connect an antenna to J2 if the sketch uses the radio.
7. Connect a USB-C data cable and select the enumerated serial port.
8. Open a basic Blink sketch and upload it.

Compile success confirms API and library compatibility; electrical operation also depends on the connected hardware and selected configuration.

5.2 First Power-up and Inspection

Use USB-C for initial power. Inspect component orientation, connector seating, and solder bridges before energizing the board. If available, use a current-limited source and monitor the 3.3 V rail.

Do not attach a battery, external `VIN`, microSD card, QWIIC peripheral, or J20 device for the first power check. Confirm the power indicator and absence of unexpected heating, then establish USB enumeration and upload behavior.

5.3 BOOT and Reset Workflow

For a normal update, the Arduino core resets the board into download mode automatically through USB Serial/JTAG. If automatic upload is unavailable:

1. Disconnect external driven signals.
2. Hold BOOT.
3. Press and release RESET, or connect USB while BOOT is held.
4. Release BOOT after the download port appears.
5. Upload the firmware using the toolchain.
6. Press RESET to run the application from external flash.

5.4 Recommended Bring-up Sequence

Order	Subsystem	What the step establishes
1	User LED	Core execution, GPIO27, and upload path

Order	Subsystem	What the step establishes
2	USB serial	Host communication and diagnostic output
3	ADC	Basic analog input path
4	RGB LED	GPIO0 timing
5	Sensor I2C	GPIO2/GPIO3, solder jumpers, and sensor addresses
6	Fuel gauge	MAX17048 voltage and state of charge
7	QWIIC I2C	GPIO1/GPIO6 and external expansion
8	PSRAM	Capacity detection and allocation
9	microSD	SPI bus, GPIO23 chip select, and filesystem
10	Wi-Fi / BLE / 802.15.4	Antenna, RF path, and network connection
11	Combined application	Memory ownership and subsystem coexistence

Testing one subsystem at a time makes pin conflicts and library configuration errors easier to isolate.

5.5 C++ Example Collection

C++ examples for the ESP32-C5 revision are planned under `software/cpp_examples/`. The recommended bring-up sequence in Section 5.4 defines the intended example coverage.

5.6 Memory Planning

Keep stacks, interrupt state, frequently accessed variables, and DMA descriptors in internal SRAM. Place large network buffers, logging queues, and historical data in PSRAM. Check every allocation and track free internal and PSRAM heap separately.

Wi-Fi, Bluetooth LE, and 802.15.4 stacks use internal memory; avoid allocating every maximum-size buffer at startup without a documented memory budget.

5.7 Storage Operation

Configure SPI on GPIO27 (SCK), GPIO7 (MOSI), and GPIO26 (MISO) with GPIO23 as the microSD chip select. Initialize the card, verify filesystem information, open only the files needed, call `flush()` after important writes, and close handles before removal or reset.

For data logging, buffer short records in RAM and write them in bounded batches rather than performing filesystem operations from an interrupt handler.

5.8 Radio, Sensor, and Storage Coexistence

The sensor bus (GPIO2/GPIO3), QWIIC bus (GPIO1/GPIO6), and SPI bus (GPIO7/GPIO26/GPIO27) do not overlap. The microSD card and J20 devices share the SPI bus and must use separate chip selects. All subsystems compete for processor time, memory, and power; radio transmission adds current peaks to the 3.3 V rail.

Use bounded update rates, buffered logging, and nonblocking network tasks. A combined application should expose health information over USB serial or RGB status rather than failing silently.

5.9 Troubleshooting Guide

Symptom	Checks
No USB device	Data-capable cable, BOOT/RESET sequence, host port, 3.3 V rail
Board does not boot	Strapping pins held low by external circuits, BOOT button
Blink does not run	GPIO27 mapping, selected board, successful upload
PSRAM unavailable	PSRAM option enabled, allocation result
Sensors not found	Sensor-bus solder jumpers, GPIO2/GPIO3, addresses
microSD fails	FAT32, GPIO23 chip select, SPI pins, card seating
Weak or no wireless link	Antenna on J2, band selection, matching network
Unstable combined app	Internal heap, PSRAM use, blocking loops, pin conflicts

Record the exact board revision, core version, library versions, compiler output, power source, and reproduction steps for every bring-up issue.

6 Mechanical Information

Hardware revision V1.2.3 uses a narrow development-board form factor with two parallel 15-position edge rows following the Arduino Nano template, USB-C at one end, and QWIIC, J20, and RF connectors for expansion.

6.1 Component-side Envelope

Board photographs for revision V1.2.3 are not yet available.

Adequate vertical and lateral clearance should be provided around the USB-C connector, pushbuttons, QWIIC and J20 connectors, RF connector, and other components. The edge pads must remain accessible when the board is mounted on headers, carrier boards, sockets, or directly soldered to another PCB.

6.2 Bottom-side Envelope

Adequate clearance must be maintained for battery connection, microSD card insertion and removal, and antenna cable routing. Enclosures and carrier boards must not cover the pressure, humidity, and light-sensor openings when those measurements are used.

6.3 Board Dimensions

A controlled dimension drawing and mounting-hole coordinates for revision V1.2.3 are not yet available. Do not derive dimensions from rendered board images.

6.4 Mechanical Integration Considerations

When integrating the UNIT PULSAR ESP32-C5 into a carrier board, fixture, or enclosure, provide sufficient clearance for:

- USB-C cable insertion and removal.
- QWIIC and J20 cable connection.
- Antenna cable and connector on J2.
- Battery connector and cable routing.
- microSD card insertion and removal.
- BOOT and RESET button access.
- Edge-pad soldering or header installation.
- Configuration solder-jumper access when required.
- Airflow and light paths to the environmental and optical sensors.

Keep metal enclosures, ground planes, and batteries away from the antenna. PCB thickness, maximum component heights, and connector insertion envelopes should be verified from the corresponding component specifications when they are critical to the final enclosure or carrier design.

7 Company Information

Item	Information
Company	UNIT Electronics
Website	UNIT Electronics
Address	Salvador 19, Cuauhtémoc, 06000 Mexico City, CDMX, Mexico

The repository and Product Wiki in the next chapter provide the product’s hardware, software, and reference documentation.

8 Reference Documentation

The following resources provide the official hardware, software, and application documentation for the UNIT PULSAR ESP32-C5.

Reference	Location
Product repository	UNIT-Electronics-MX/unit_pulsar_esp32_c5
Hardware reference	hardware/README.md
Technical wiki	PULSAR ESP32-C5 Wiki
Board schematic	Hardware V1.2.3 schematic
ESP32-C5 component datasheet	Espressif ESP32-C5 Series Datasheet
Arduino core	Espressif Arduino-ESP32

The UNIT Electronics Wiki provides application-oriented documentation, setup procedures, peripheral usage examples, and software workflows for the board.

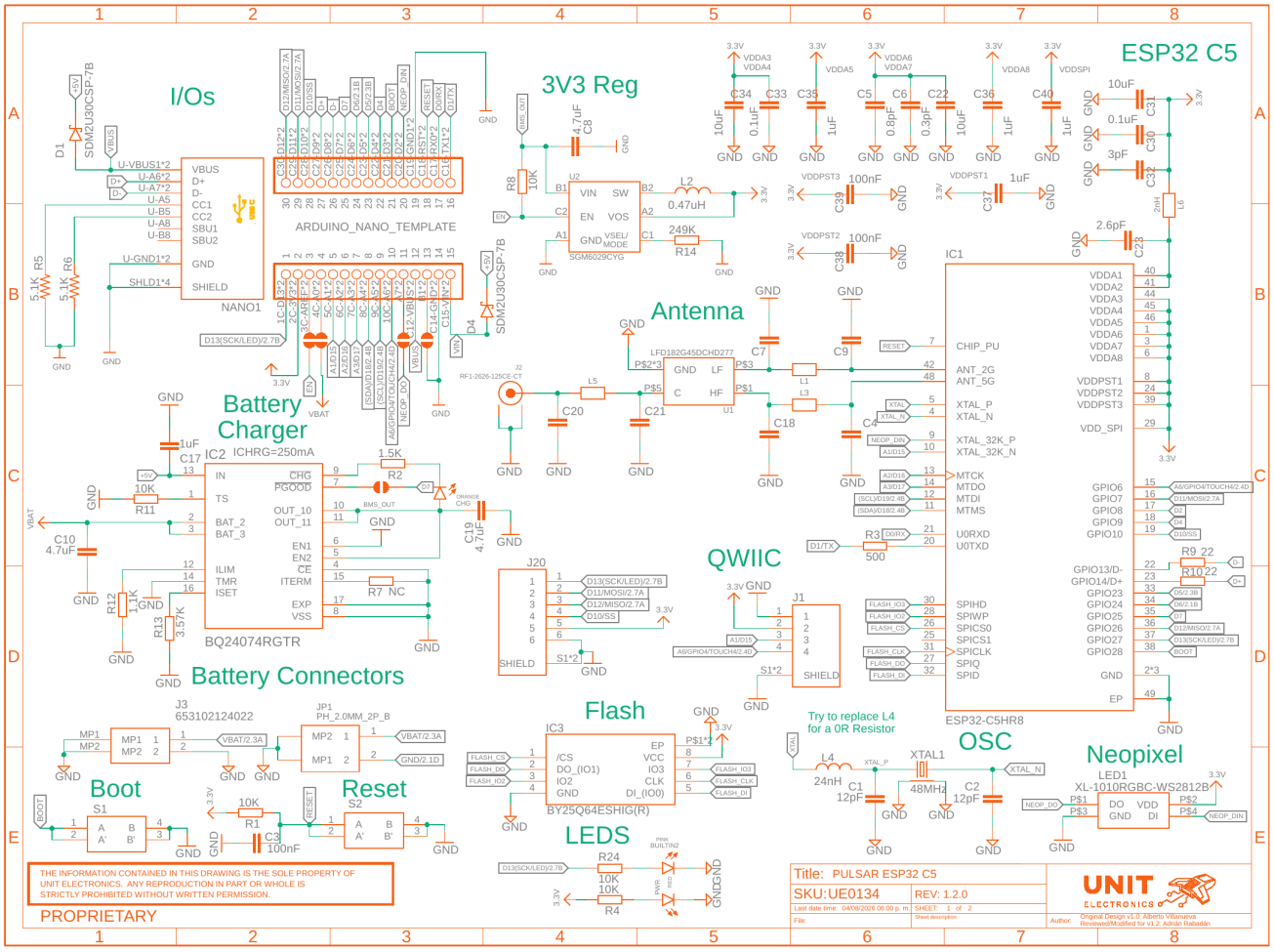
The hardware schematic and this Product Reference define the board-level implementation, connections, power architecture, interfaces, and hardware characteristics. ESP32-C5HR8 device-level electrical characteristics and processor specifications should be referenced from the official Espressif ESP32-C5 datasheet.

9 Appendix

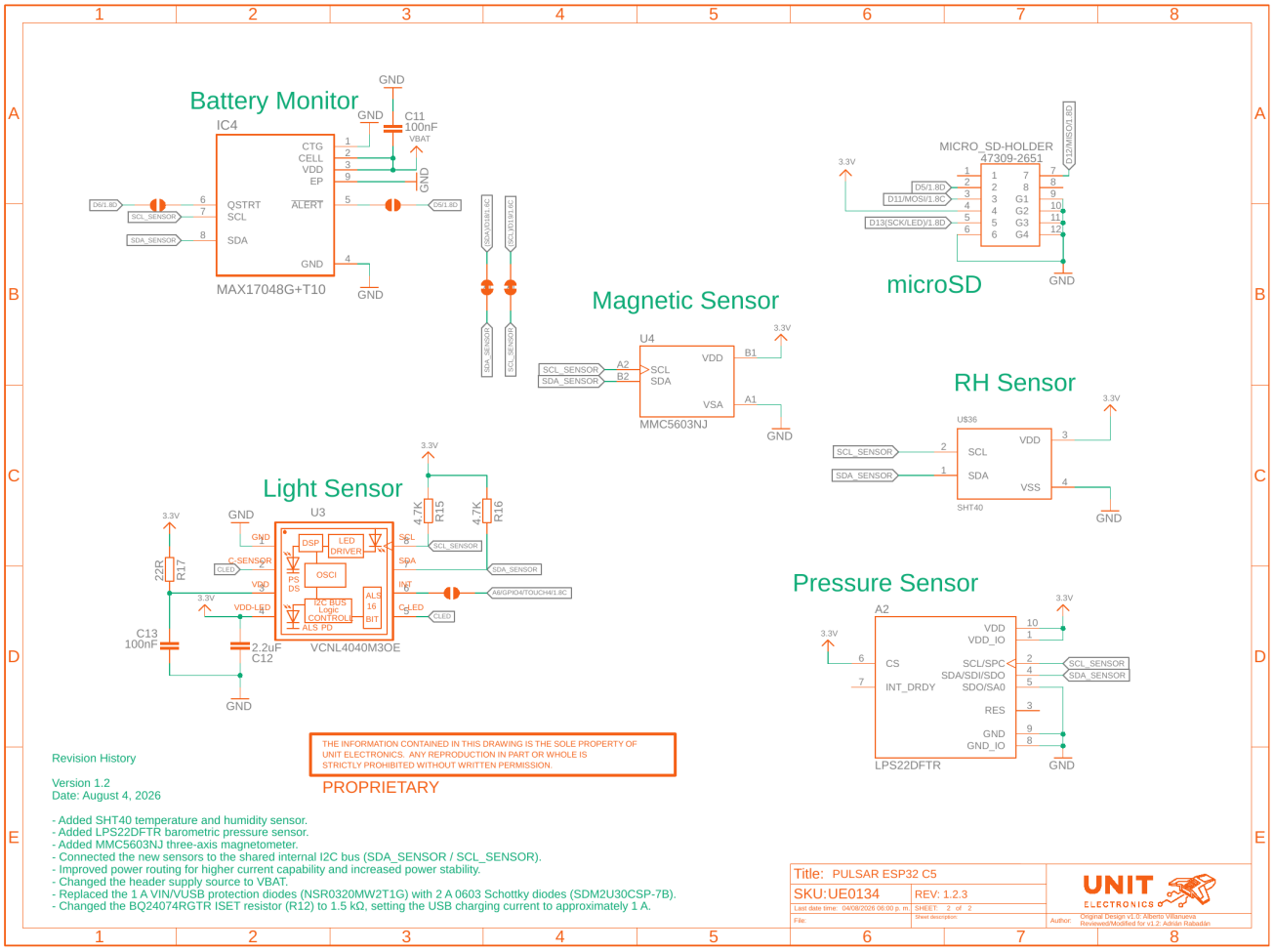
9.1 Schematic

The following two sheets are rendered from `unit_sch_v_1_2_3_ue0134_pulsar_esp32_c5.pdf`. The original PDF remains the authoritative electrical schematic for hardware revision V1.2.3.

9.1.1 I/Os, USB-C, Charger, 3.3 V Regulator, Antenna, ESP32-C5, Flash, and LEDs



9.1.2 Battery Monitor, microSD, and Sensors



9.2 Technical References

The documentation set for the UNIT PULSAR ESP32-C5 is organized around the following technical sources:

1. The hardware V1.2.3 schematic defines board connectivity, component designators, power architecture, and electrical implementation.
2. The bill of materials defines the assembled part numbers.
3. The official Espressif ESP32-C5 datasheet defines device-level electrical characteristics, processor capabilities, GPIO behavior, strapping pins, ADC operation, and radio functionality.
4. The UNIT Electronics Wiki provides setup procedures, peripheral usage documentation, application examples, and software workflows.

9.3 Design and Application Notes

The following characteristics depend on the final application, firmware configuration, connected peripherals, or mechanical integration:

- Total board current consumption varies with CPU activity, radio mode and transmit power, enabled peripherals, RGB LED brightness, microSD activity, and external loads.
- Available current for external 3.3 V devices depends on the portion of the SGM6029 1 A output capability consumed by the board itself.
- Radiated RF performance depends on the antenna, cable, enclosure, and the populated matching network.
- Environmental-sensor accuracy depends on self-heating, airflow, and enclosure design.
- Thermal performance and maximum component temperatures depend on workload, ambient conditions, airflow, enclosure design, and external loading.

9.4 Document Control

Field	Value
Product	UNIT PULSAR ESP32-C5
SKU	UE0134
Product family	UNIT DevLab ecosystem
Hardware revision	V1.2.3
Product Reference	Version 0.1.0
Publication date	2026-09-24

9.5 Source Notes

- The schematic title block shows `REV: 1.2.0` on sheet 1 and `REV: 1.2.3` on sheet 2; the schematic file name uses V1.2.3, and the revision history is labeled Version 1.2. This Product Reference uses V1.2.3.
- The schematic names the humidity sensor `SHT40` and the pressure sensor `LPS22DFTR`; the bill of materials specifies `FHT40-DD-TR` and `LPS22HBTR`. This Product Reference follows the bill of materials for part numbers.
- The schematic revision history states that R12 was changed to 1.5 kΩ for an approximately 1 A USB charging current. The schematic and bill of materials show R12 = 1.1 kΩ on `ILIM` (input current limit) and R13 = 3.57 kΩ on `ISET`, with the annotation `ICHRG=250mA`. This Product Reference uses the schematic and bill-of-materials values.
- The J3 part number is `653102124022` in the schematic and `653102131822` in the bill of materials.
- GPIO8 carries the net label D2 at IC1, but no header connection is shown for that net. Header position D2 connects to GPIO0 (`NEOP_DIN`).
- The net connected to header position 10 and QWIIC pin 4 is named `A6/GPIO4/TOUCH4`, but it connects to IC1 pin 15 (GPIO6). The ESP32-C5 has no touch sensor. This Product Reference uses GPIO6.
- The label D21 for header position 10 is taken from the board pinout artwork; the schematic shows only A6.
- The ESP32-C5 does not provide an SD host controller; the microSD socket uses SPI mode and DAT1/DAT2 are not connected.

9.6 Hardware Errata

No hardware errata have been recorded for revision V1.2.3.